

Department of Information Engineering
MSc in Computer Engineering (a.a. 2022/2023)
Computer Architecture (Prof. Cosimo Antonio Prete)

Introduction To Quantum Computing

Luciano Lenzini
Full Professor
Department of Information Engineering
School of Engineering
University of Pisa, Italy
e-mail: luciano.lenzini@dii.unipi.it
<http://www.iet.unipi.it/~lenzini/>
<http://www.originiinternetitalia.it/it/>



Linear Algebra

- Given a matrix A

$$A = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \text{ where } a, b, c, \text{ and } d \text{ are complex numbers}$$

- The *adjoint* or *Hermitian conjugate* $A^\dagger$ of A is the *transpose* of its *complex conjugate*

$$A^\dagger = \begin{bmatrix} a & c \\ b & d \end{bmatrix}^\dagger = \left(\begin{bmatrix} a & c \\ b & d \end{bmatrix}^* \right)^T = \left(\begin{bmatrix} a^* & c^* \\ b^* & d^* \end{bmatrix} \right)^T = \begin{bmatrix} a^* & b^* \\ c^* & d^* \end{bmatrix} \quad \Rightarrow \quad [A^\dagger]_{ij} = [A^*]_{ji}$$

- Sometimes $A^\dagger$ is pronounced “A dagger”

Linear Algebra

- We would have achieved the same result if we had *first transposed* the matrix A and then taken the *conjugate of the elements*

$$A^\dagger = \begin{bmatrix} a & c \\ b & d \end{bmatrix}^\dagger = \left(\begin{bmatrix} a & c \\ b & d \end{bmatrix}^T \right)^* = \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right)^* = \begin{bmatrix} a^* & b^* \\ c^* & d^* \end{bmatrix}$$

- Therefore, the generic element of $A^\dagger$ can be written as

$$[A^\dagger]_{ij} = \left([A]_{ji} \right)^*$$

Linear Algebra

- By exploiting the result $\left[A^\dagger\right]_{ij} = \left([A_{ji}]\right)^*$ we can prove

$$(AB)^\dagger = B^\dagger A^\dagger$$

- Proof

$$\left([AB]^\dagger\right)_{ij} = \left([AB]_{ji}\right)^* = \sum_k \left([A]_{jk} [B]_{ki}\right)^* = \sum_k \left([A]_{jk}\right)^* \left([B]_{ki}\right)^*$$

$$= \sum_k \left[B^\dagger\right]_{ik} \left[A^\dagger\right]_{kj} = \left[B^\dagger A^\dagger\right]_{ij}$$



Linear Algebra

It is easy to prove that $(A^\dagger)^\dagger = A$

- Proof

$$\left(A^\dagger\right)^\dagger = \begin{bmatrix} a^* & b^* \\ c^* & d^* \end{bmatrix}^\dagger = \left(\begin{bmatrix} a^* & b^* \\ c^* & d^* \end{bmatrix}^*\right)^T = \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right)^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix} = A$$

From the dagger definition of a matrix, it follows

$$|\psi\rangle^\dagger = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}^\dagger = \left(\begin{bmatrix} \alpha \\ \beta \end{bmatrix}^*\right)^T = \left(\begin{bmatrix} \alpha^* \\ \beta^* \end{bmatrix}\right)^T = (\alpha^* \quad \beta^*) = \langle\psi| \quad \longrightarrow \quad \boxed{|\psi\rangle^\dagger = \langle\psi|}$$

Quantum Computation

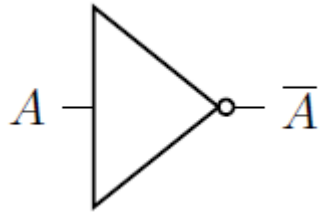
Quantum Computation

- A classical computer is built from an *electrical circuit* containing *wires* and *logic gates*
- The *wires* are used to carry information around the circuit, while the *logic gates* perform manipulations of the information, converting it from one form to another
- Similar to a classical computer, a quantum computer is built from a *quantum circuit* containing *wires* and elementary *quantum gates* to carry around and manipulate the quantum information
- Thus, quantum gates act on qubits, like logic gates act on bits
- Specifically, **a quantum gate transforms the state of a qubit into other states**

Single Qubit Gates

Classical Computer

- Consider, for example, the NOT classical single bit logic gate, whose operation is defined by its *truth table*



Input bit A	Output bit $\bar{A}$
0	1
1	0

in which $0 \rightarrow 1$ and $1 \rightarrow 0$, that is, the 0 and 1 states are interchanged

Single Qubit Gates

Question: can an analogous quantum gate for qubits be defined?

- Imagine that we have a quantum gate which takes the state $|0\rangle$ to the state $|1\rangle$ and vice versa
- We denote a single-qubit gate with a box containing the label (straddling the line) representing the operation carried out by the gate



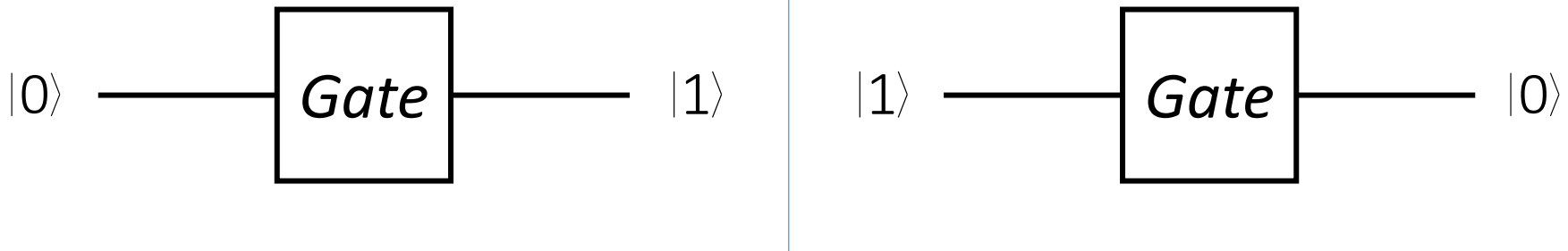
Single Qubit Gates

- Let's call *Gate* the operator which flips the state of a qubit from state $|0\rangle$ to state $|1\rangle$ and vice versa

$$\textit{Gate}|0\rangle = |1\rangle$$

$$\textit{Gate}|1\rangle = |0\rangle$$

Such a quantum operator would obviously be a good candidate for a quantum analogue to the NOT gate



Single Qubit Gates

- However, the specification given in the previous slide is not enough!
- **Question:** why is that?
- **Answer:** because specifying the action of the gate on the states $|0\rangle$ and $|1\rangle$ does not tell us what happens to superpositions of the states $|0\rangle$ and $|1\rangle$

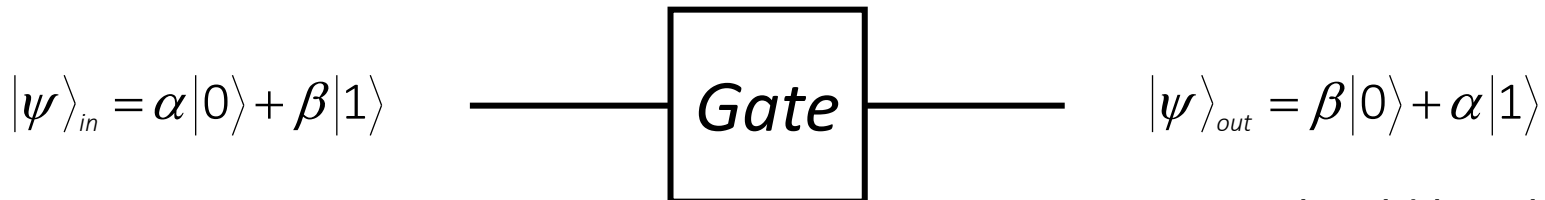
Single Qubit Gates

- In fact, the quantum gate acts *linearly*, that is, operating on a superposition it would do the following

$$\begin{aligned} |\psi\rangle_{out} &= Gate|\psi\rangle_{in} = Gate(\alpha|0\rangle + \beta|1\rangle) = \alpha(Gate|0\rangle) + \beta(Gate|1\rangle) \\ &= \alpha|1\rangle + \beta|0\rangle = \beta|0\rangle + \alpha|1\rangle \end{aligned}$$



$$|\psi\rangle_{out} = \beta|0\rangle + \alpha|1\rangle$$



As it should be, the output state is normalized.

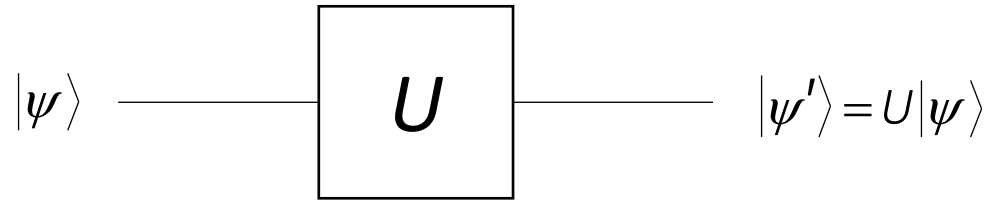
NOTE

- Why the quantum gate *acts linearly* and not in some *nonlinear* fashion is a very interesting question, and the answer is not at all obvious
- It turns out that this *linear behavior* is a general property of quantum mechanics, and *very well motivated empirically*; moreover, nonlinear behavior can lead to apparent paradoxes such *as faster-than-light communication*, and *violations of the second laws of thermodynamics*

Single Qubit Gates

- Before doing an in-depth analysis of specific one-qubit quantum gates we extend the previous result to the case of the most general one-qubit gate
- As we said before a quantum gate *transforms* the state of a qubit into other states

Single Qubit Gates



- Let's denote by U such a transformation which generally turns $|0\rangle$ and $|1\rangle$ into superpositions of $|0\rangle$ and $|1\rangle$

$$\begin{aligned} U|0\rangle &= a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix} \\ U|1\rangle &= c|0\rangle + d|1\rangle = \begin{bmatrix} c \\ d \end{bmatrix} \end{aligned}$$

We can arrange the resulting amplitudes side-by-side, resulting in the following 2x2 matrix



$$U = \begin{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} & \begin{bmatrix} c \\ d \end{bmatrix} \end{bmatrix} = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

Single Qubit Gates

$$U|0\rangle = a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$
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$$U = \begin{matrix} & |0\rangle & |1\rangle \\ \begin{matrix} \langle 0| \\ \langle 1| \end{matrix} & \begin{bmatrix} \langle 0|U|0\rangle & \langle 0|U|1\rangle \\ \langle 1|U|0\rangle & \langle 1|U|1\rangle \end{bmatrix} \end{matrix} = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

Single Qubit Gates

- Plugging this matrix into $U|0\rangle$ and $U|1\rangle$

$$U|0\rangle = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$U|1\rangle = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c \\ d \end{bmatrix}$$

From the previous slide

$$U|0\rangle = a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$U|1\rangle = c|0\rangle + d|1\rangle = \begin{bmatrix} c \\ d \end{bmatrix}$$

Single Qubit Gates

- Remember that the quantum gate (or U) acts *linearly*, meaning that if a qubit is in the state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

From the previous slide

$$\begin{aligned} U|0\rangle &= a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix} \\ U|1\rangle &= c|0\rangle + d|1\rangle = \begin{bmatrix} c \\ d \end{bmatrix} \end{aligned}$$

then applying U transforms this to

$$U|\psi\rangle = \alpha U|0\rangle + \beta U|1\rangle = \alpha \begin{bmatrix} a \\ b \end{bmatrix} + \beta \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a\alpha \\ b\alpha \end{bmatrix} + \begin{bmatrix} c\beta \\ d\beta \end{bmatrix} = \begin{bmatrix} a\alpha + c\beta \\ b\alpha + d\beta \end{bmatrix}$$

Single Qubit Gates

From the above slide we have

$$U|\psi\rangle = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} a\alpha + c\beta \\ b\alpha + d\beta \end{bmatrix} = (a\alpha + c\beta) \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (b\alpha + d\beta) \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

or

$$U|\psi\rangle = (a\alpha + c\beta)|0\rangle + (b\alpha + d\beta)|1\rangle$$

Assuming the original state was normalized, i.e. $|\alpha|^2 + |\beta|^2 = 1$, this must also be true for the quantum state after the gate has acted

Single Qubit Gates

- Of course, the matrix must ensure that the total probability remains 1, so in the previous example, we must have

$$|a\alpha + c\beta|^2 + |b\alpha + d\beta|^2 = 1$$

- This yields the following point: *quantum gates are (represented by) matrices that keep the total probability equal to 1*

$$U|\psi\rangle = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} a\alpha + c\beta \\ b\alpha + d\beta \end{bmatrix}$$

From the previous slide



Example

- For example, consider a quantum gate that performs the following transformation:

$$\begin{aligned} U|0\rangle &= \frac{\sqrt{2}-i}{2}|0\rangle - \frac{1}{2}|1\rangle = \begin{bmatrix} \frac{\sqrt{2}-i}{2} \\ -\frac{1}{2} \end{bmatrix}, \\ U|1\rangle &= \frac{1}{2}|0\rangle + \frac{\sqrt{2}+i}{2}|1\rangle = \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{2}+i}{2} \end{bmatrix} \end{aligned} \quad \longrightarrow \quad U = \begin{bmatrix} \frac{\sqrt{2}-i}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{2}+i}{2} \end{bmatrix}$$

Example

A quantum gate must be *linear*, meaning we can distribute it across superpositions:

$$\begin{aligned} U(\alpha|0\rangle + \beta|1\rangle) &= \alpha U|0\rangle + \beta U|1\rangle \\ &= \alpha \left(\frac{\sqrt{2}-i}{2}|0\rangle - \frac{1}{2}|1\rangle \right) + \beta \left(\frac{1}{2}|0\rangle + \frac{\sqrt{2}+i}{2}|1\rangle \right) \\ &= \left(\alpha \frac{\sqrt{2}-i}{2} + \beta \frac{1}{2} \right) |0\rangle + \left(-\alpha \frac{1}{2} + \beta \frac{\sqrt{2}+i}{2} \right) |1\rangle. \end{aligned}$$

Example

For this to be a valid quantum gate, the total probability must remain 1. Assuming the original state was normalized, i.e., $|\alpha|^2 + |\beta|^2 = 1$, we can calculate the total probability by summing the norm-square of each amplitude to see if it is still 1:

$$\begin{aligned} & \left| \alpha \frac{\sqrt{2}-i}{2} + \beta \frac{1}{2} \right|^2 + \left| -\alpha \frac{1}{2} + \beta \frac{\sqrt{2}+i}{2} \right|^2 \\ &= \left(\alpha \frac{\sqrt{2}-i}{2} + \beta \frac{1}{2} \right) \left(\alpha^* \frac{\sqrt{2}+i}{2} + \beta^* \frac{1}{2} \right) \\ & \quad + \left(-\alpha \frac{1}{2} + \beta \frac{\sqrt{2}+i}{2} \right) \left(-\alpha^* \frac{1}{2} + \beta^* \frac{\sqrt{2}-i}{2} \right) \\ &= |\alpha|^2 \frac{(\sqrt{2}-i)(\sqrt{2}+i)}{4} + \alpha\beta^* \frac{\sqrt{2}-i}{2} + \beta\alpha^* \frac{\sqrt{2}+i}{4} + |\beta|^2 \frac{1}{4} \\ & \quad + |\alpha|^2 \frac{1}{4} - \alpha\beta^* \frac{\sqrt{2}-i}{4} - \beta\alpha^* \frac{\sqrt{2}+i}{4} + |\beta|^2 \frac{(\sqrt{2}+i)(\sqrt{2}-i)}{4} \\ &= |\alpha|^2 \frac{3}{4} + |\beta|^2 \frac{1}{4} + |\alpha|^2 \frac{1}{4} + |\beta|^2 \frac{3}{4} \\ &= |\alpha|^2 + |\beta|^2 \\ &= 1. \end{aligned}$$



So, U is a valid quantum gate. Then,

Quantum gates are linear maps that keep the total probability equal to 1.

Unitarity

- Let's return to the previous example where

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

and

$$U|\psi\rangle = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} a\alpha + c\beta \\ b\alpha + d\beta \end{bmatrix}$$

- We see that $U|\psi\rangle$ is a column vector, so we can also write it as a ket $|\psi\rangle$

$$U|\psi\rangle = |\psi\rangle \quad \text{where} \quad |\psi\rangle = \begin{bmatrix} a\alpha + c\beta \\ b\alpha + d\beta \end{bmatrix}$$

Unitarity

- Now, consider the *conjugate transpose* of $U|\psi\rangle$

$$\begin{aligned}
 \langle\psi| &= [(a\alpha + c\beta)^* \quad (b\alpha + d\beta)^*] = [a^*\alpha^* + c^*\beta^* \quad b^*\alpha^* + d^*\beta^*] \\
 &= [\alpha^* \quad \beta^*] \begin{bmatrix} a^* & b^* \\ c^* & d^* \end{bmatrix} = [\alpha^* \quad \beta^*] \begin{bmatrix} a & b \\ c & d \end{bmatrix}^* \\
 &= [\alpha^* \quad \beta^*] \left(\begin{bmatrix} a & c \\ b & d \end{bmatrix}^* \right)^T = \underbrace{[\alpha^* \quad \beta^*]}_{\langle\psi|} \underbrace{\begin{bmatrix} a & c \\ b & d \end{bmatrix}^\dagger}_U = \langle\psi|U^\dagger
 \end{aligned}$$

- To summarize

$$|\psi\rangle = U|\psi\rangle \longrightarrow \langle\psi| = \langle\psi|U^\dagger$$

Unitarity

- We can reach the result $\langle \psi | = \langle \psi | U^\dagger$ by considering that

$$\langle \psi | = (U |\psi\rangle)^\dagger = |\psi\rangle^\dagger U^\dagger = \langle \psi | U^\dagger$$

$$|\psi\rangle^\dagger = \langle \psi |$$

$$(AB)^\dagger = B^\dagger A^\dagger$$

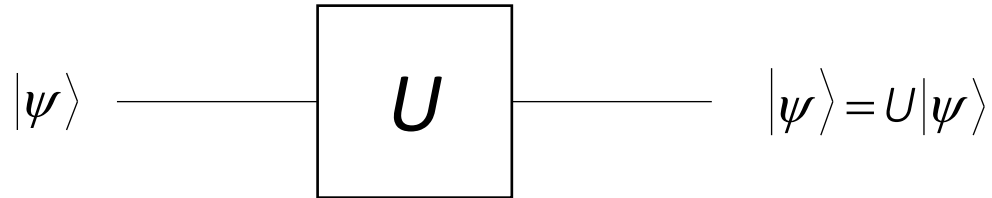
Unitarity

- Using this, we can come up with an easy way to determine whether a matrix keeps the total probability equal to 1
- Consider a quantum gate (matrix) U
- If it acts on $|\psi\rangle$, we have

$$U|\psi\rangle = |\psi\rangle$$

- For U to be a quantum gate, $|\psi\rangle$ must be normalized, that is, the inner product of $|\psi\rangle$ with itself must equal 1

Unitarity



$$\langle\psi|\psi\rangle=1 \rightarrow$$

$$\langle\psi|U^\dagger U|\psi\rangle=1, \quad \forall U \rightarrow U^\dagger U = I$$

A matrix that satisfies this property $U^\dagger U = I$ and $UU^\dagger = I$ is called *unitary*

Quantum gates are *unitary matrices*, and *unitary matrices* are quantum gates

Unitarity

This is why we typically use U to denote a quantum gate. It stands for unitary. As an example application of this, is the following matrix a quantum gate?

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$$

We can just check whether it is unitary, so whether $U^\dagger U = I$ or not.

$$U^\dagger U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} = \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} \neq I.$$

So no, it's not a quantum gate.

Reversibility

- A matrix M is *reversible* or *invertible* if there exists a matrix denoted M^{-1} such that

$$MM^{-1} = M^{-1}M = I$$

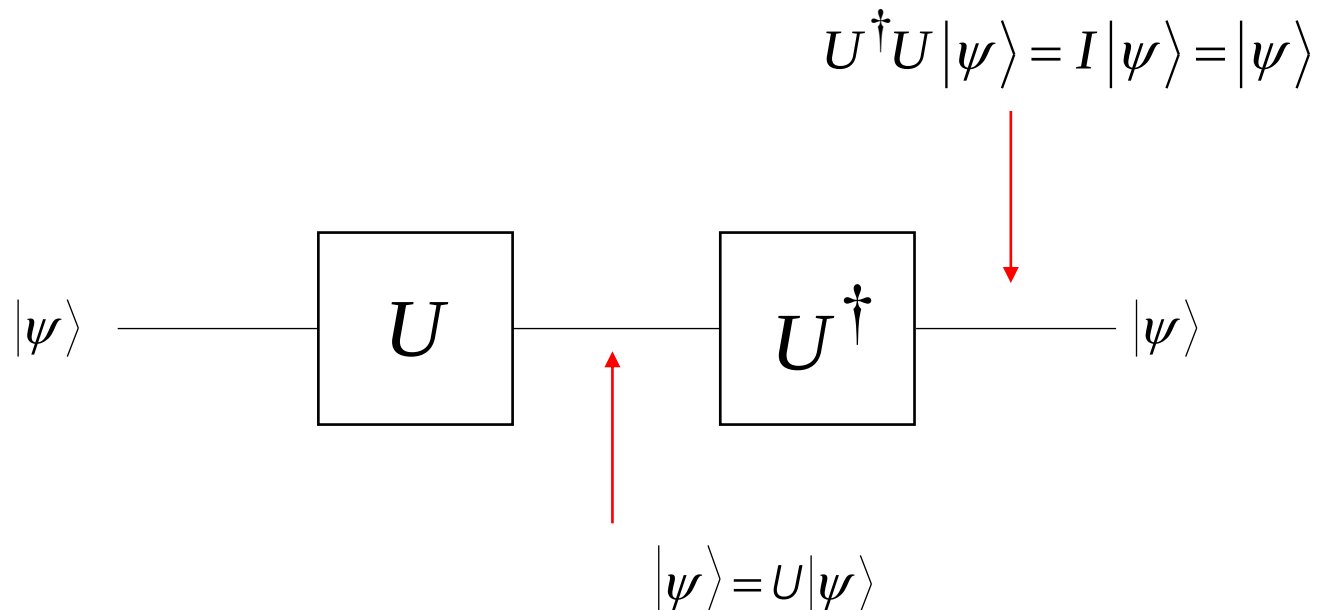
- Now, since a quantum gate U must be unitary, i.e.

$$UU^\dagger = U^\dagger U = I$$

it follows that the inverse of the unitary matrix U is simply $U^{-1} = U^\dagger$

Reversibility

- As a consequence, a quantum gate U is *always reversible*
- If we have a qubit and we applied a quantum gate U , we can undo the operation by applying $U^\dagger$



Outer Product

- Given two states

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad |\phi\rangle = \gamma|0\rangle + \delta|1\rangle$$

we can define their *outer state* $|\psi\rangle\langle\phi|$

$$|\psi\rangle\langle\phi| = (|\psi\rangle) \cdot (\langle\phi|) = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \begin{bmatrix} \gamma^* & \delta^* \end{bmatrix} = \begin{bmatrix} \alpha\gamma^* & \alpha\delta^* \\ \beta\gamma^* & \beta\delta^* \end{bmatrix}$$

- So, the outer product of two qubit states is a 2X2 matrix

Outer Product

- Let's prove that

$$|\phi\rangle\langle\psi| = (|\psi\rangle\langle\phi|)^{\dagger}$$

- Proof

$$(|\psi\rangle\langle\phi|)^{\dagger} = (\langle\phi|)^{\dagger} (|\psi\rangle)^{\dagger} = |\phi\rangle\langle\psi|$$

$$(AB)^{\dagger} = B^{\dagger}A^{\dagger}$$


Completeness Relation

- Recall that for any orthonormal basis $\{|a\rangle, |b\rangle\}$, the state of a qubit can be written as

$$|\psi\rangle = \alpha|a\rangle + \beta|b\rangle \quad \text{where} \quad \alpha = \langle a|\psi\rangle \quad \beta = \langle b|\psi\rangle$$

- Substituting these values

$$|\psi\rangle = \underbrace{\langle a|\psi\rangle}_{\text{scalar}}|a\rangle + \underbrace{\langle b|\psi\rangle}_{\text{scalar}}|b\rangle$$

Completeness Relation

- As show above, the inner products are just scalars/numbers, so instead of multiply them onto the vectors $|a\rangle$ and $|b\rangle$ on the left, we can equivalently multiply them on the right

$$|\psi\rangle = |a\rangle \underbrace{\langle a|\psi\rangle}_{\text{scalar}} + |b\rangle \underbrace{\langle b|\psi\rangle}_{\text{scalar}}$$

- Both of these terms are a *ket* times a *bra* times a *ket*

Completeness Relation

- To make this more clear, we can write them as

$$|\psi\rangle = |a\rangle\langle a|\psi\rangle + |b\rangle\langle b|\psi\rangle$$

- Now, notice we have two outer products, $|a\rangle\langle a|$ and $|b\rangle\langle b|$
- Since they are both multiplying $|\psi\rangle$, we can factor to get

$$|\psi\rangle = (|a\rangle\langle a| + |b\rangle\langle b|)|\psi\rangle$$

Completeness Relation

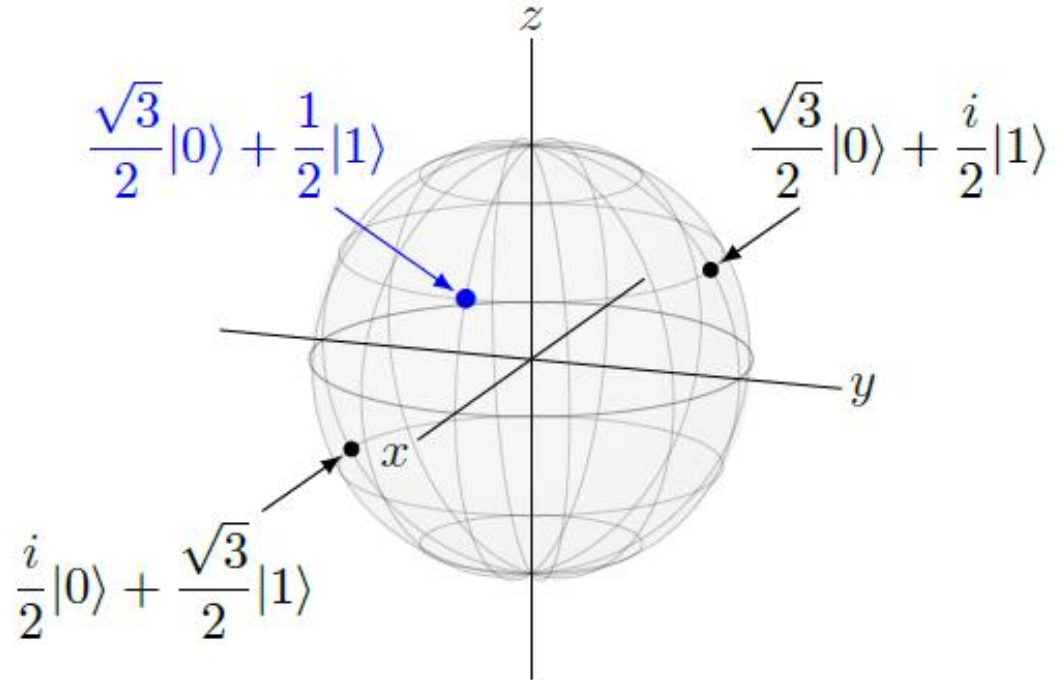
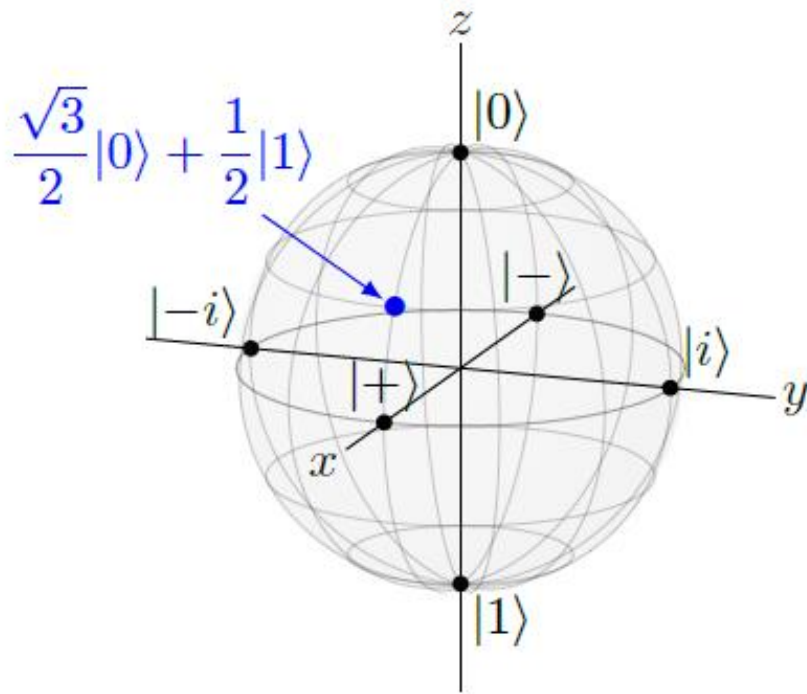
- For this to be true for all $|\psi\rangle$, we must have

$$|a\rangle\langle a| + |b\rangle\langle b| = I$$

- This is called the *completeness relation*

Completeness Relation

- We say $\{|a\rangle, |b\rangle\}$ forms a *complete orthonormal basis*
- All the bases we have discussed (any two states on opposite sides on the Bloch sphere) are complete



Let's Play with the Outer Product

- Let V be a vector space associated with a single-qubit system
- The outer products $|i\rangle\langle j|$, $i, j \in \{0,1\}$ with respect to the standard basis $\{|0\rangle, |1\rangle\}$ are

$$|0\rangle\langle 0| = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$|0\rangle\langle 1| = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$|1\rangle\langle 0| = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$|1\rangle\langle 1| = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Completeness Relation

- Based on the above results, the *completeness relation* can be easily checked

$$|0\rangle\langle 0| + |1\rangle\langle 1| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

- Since $\{|+\rangle, |-\rangle\}$ is an *orthonormal basis*, following the same approach as before we could show that

$$|+\rangle\langle +| + |-\rangle\langle -| = I$$

i.e. $\{|+\rangle, |-\rangle\}$ is a *complete orthonormal basis*

Completeness Relation

- Let's prove that $\{|+\rangle, |-\rangle\}$ is a *complete orthonormal basis*, i.e.

$$|+\rangle\langle+| + |-\rangle\langle-| = I$$

- Proof

$$|+\rangle\langle+| = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \frac{1}{\sqrt{2}} [1 \quad 1] = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad |-\rangle\langle-| = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \frac{1}{\sqrt{2}} [1 \quad -1] = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad \rightarrow$$

$$|+\rangle\langle+| + |-\rangle\langle-| = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Completeness Relation

- On the other hand, $\{|0\rangle, |+\rangle\}$ is not a *complete orthonormal basis* due to

$$|0\rangle\langle 0| + |+\rangle\langle +| \neq I$$

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- Proof

$$|0\rangle\langle 0| = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad |+\rangle\langle +| = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \longrightarrow$$

$$|0\rangle\langle 0| + |+\rangle\langle +| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

One-Qubit Quantum Gates

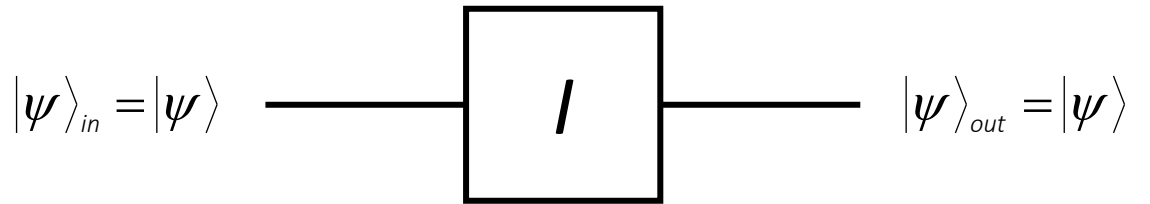
Reference

To find out more about quantum gates, I recommend you read the paper

Quantum computers: registers, gates and algorithm
by Paul Isaac Hagouel

Identity Gate (I)

- The *identity gate* turns $|0\rangle$ into $|0\rangle$ and $|1\rangle$ into $|1\rangle$, hence doing nothing
- The *input* state is placed on the *left* of the gate symbol and the *output* state on the *right*



$$\begin{aligned} I|0\rangle &= |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ I|1\rangle &= |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned} \quad \Rightarrow \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \equiv |0\rangle\langle 0| + |1\rangle\langle 1|$$

The Pauli X -Gate, or NOT Gate

- This gate, turns $|0\rangle$ into $|1\rangle$, and $|1\rangle$ into $|0\rangle$

$$X|0\rangle = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



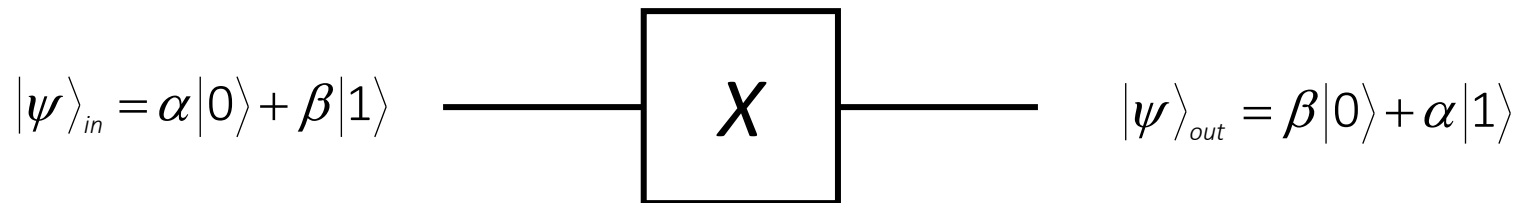
$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \equiv |0\rangle\langle 1| + |1\rangle\langle 0|$$

$$X|1\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

The Pauli X-Gate, or NOT Gate

- When acting on $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $|\alpha|^2 + |\beta|^2 = 1$, X swaps the amplitudes α and β
- The X operator is sometimes called the *bit flip operator*, because it “flips” the CBS amplitudes, $\alpha \leftrightarrow \beta$

$$|\psi_{out}\rangle = X \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}$$



The Pauli X -Gate, or NOT Gate

- Since

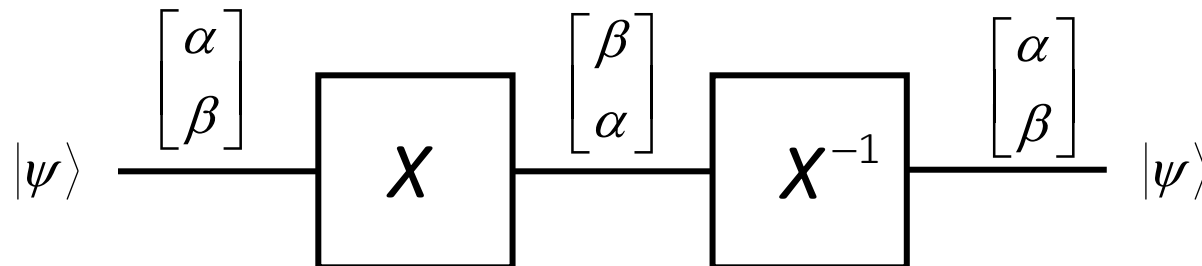
$$X = X^\dagger \quad \longrightarrow \quad XX^\dagger = X^\dagger X = XX = X^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

- It turns out that the matrix X describing the single qubit gate is **unitary**

The Pauli X-Gate, or NOT Gate

- Unitary quantum gates are *always invertible*, since the inverse of a unitary matrix is also a unitary matrix, and thus a quantum gate can always be inverted by another quantum gate

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \rightarrow X^\dagger = X^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = X$$



The Pauli Z-Gate or Phase Flip Gate

- This gate, keeps $|0\rangle$ as $|0\rangle$, and turns $|1\rangle$ into $-|1\rangle$

$$\begin{aligned} Z|0\rangle &= |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ Z|1\rangle &= -|1\rangle = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \end{aligned} \quad \Rightarrow \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \equiv |0\rangle\langle 0| - |1\rangle\langle 1|$$

- Z is unitary and thus reversible

$$ZZ^\dagger = Z^\dagger Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

The Pauli Z-Gate or Phase Flip Gate

When acting on $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $|\alpha|^2 + |\beta|^2 = 1$

$$Z \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ -\beta \end{bmatrix} = \alpha|0\rangle - \beta|1\rangle$$

Z leaves $|0\rangle$ unchanged, and flips the sign of $|1\rangle$ to give $-|1\rangle$

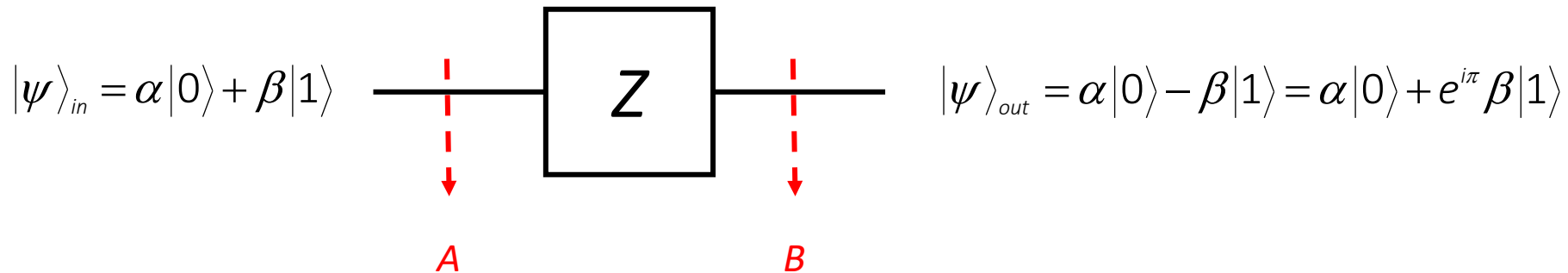
$$|\psi\rangle_{in} = \alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{Z} \longrightarrow |\psi\rangle_{out} = \alpha|0\rangle - \beta|1\rangle = \alpha|0\rangle + e^{i\pi}\beta|1\rangle$$

$-1 = e^{i\pi}$



The Pauli Z-Gate or Phase Flip Gate

- Z is also called a *phase flip*, because it changes the sign of the amplitude of the state $|1\rangle$



- The circuit produces the A -to- B transition $\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \rightarrow \begin{bmatrix} \alpha \\ -\beta \end{bmatrix}$ yielding the same probabilities, $|\alpha|^2$ and $|\beta|^2$, at both access point

The Pauli Y-Gate

- This gate, turns $|0\rangle$ into $i|1\rangle$, and $|1\rangle$ into $-i|0\rangle$

$$Y|0\rangle = i|1\rangle = \begin{bmatrix} 0 \\ i \end{bmatrix}$$

$$Y|1\rangle = -i|0\rangle = \begin{bmatrix} -i \\ 0 \end{bmatrix}$$



$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = -i|0\rangle\langle 1| + i|1\rangle\langle 0|$$

The Pauli Y-Gate

- Y is *unitary* and thus *reversible*

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad \longrightarrow \quad Y^\dagger = \left(\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}^* \right)^T = \left(\begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} \right)^T = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = Y$$

$$Y^\dagger = Y \quad \longrightarrow$$

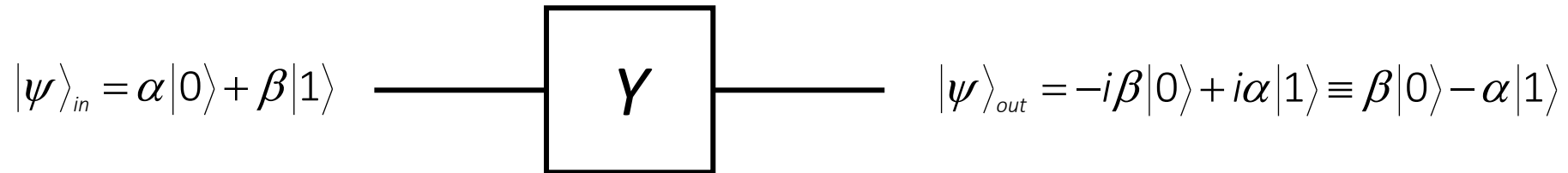
- Thus

$$YY^\dagger = Y^\dagger Y = YY = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

The Pauli Y-Gate

- When acting on $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $|\alpha|^2 + |\beta|^2 = 1$

$$Y \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -i\beta \\ i\alpha \end{bmatrix} = -i \begin{bmatrix} \beta \\ -\alpha \end{bmatrix} \equiv \beta|0\rangle - \alpha|1\rangle$$



- Although Y has no official name, it could be called the *bit-and-phase flip*, because it *flips* both the *bits* and *the relative phase*, simultaneously

The Phase-Shift Gates, S and T

- We have already seen that the phase-flip gate, Z , when operates on a qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ *flips* the relative phase, i.e. *shifts* it, by π radians

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \rightarrow \begin{bmatrix} \alpha \\ -\beta \end{bmatrix} = \begin{bmatrix} \alpha \\ e^{i\pi} \beta \end{bmatrix}$$

- There are two other common shift amounts, $\pi/2$ (the S operator) and $\pi/4$ (the T operator)

The Phase S-Gate

- This gate, keeps $|0\rangle$ as $|0\rangle$, and turns $|1\rangle$ into $i|1\rangle$

$$\begin{aligned} S|0\rangle &= |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ S|1\rangle &= i|1\rangle = \begin{bmatrix} 0 \\ i \end{bmatrix} \end{aligned} \quad \Rightarrow \quad S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \equiv |0\rangle\langle 0| + i|1\rangle\langle 1|$$

The Phase S-Gate

- S is unitary

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \quad \longrightarrow \quad S^\dagger = \left(\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}^* \right)^T = \left(\begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix} \right)^T = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}$$

$$SS^\dagger = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

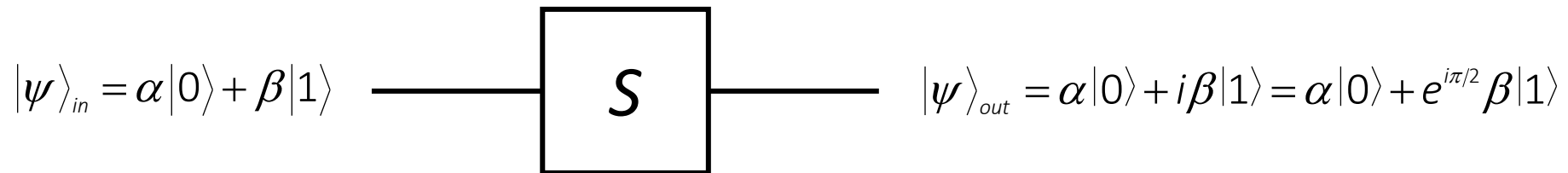
$$S^\dagger S = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$\longrightarrow SS^\dagger = S^\dagger S = I$$

The Phase S-Gate

- When acting on $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $|\alpha|^2 + |\beta|^2 = 1$

$$S \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ i\beta \end{bmatrix} = \alpha|0\rangle + i\beta|1\rangle = \alpha|0\rangle + e^{i\pi/2}\beta|1\rangle$$



The $\pi/8$ or T -Gate

- This gate, keeps $|0\rangle$ as $|0\rangle$, and turns $|1\rangle$ into $e^{i\pi/4}|1\rangle$

$$T|0\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} \equiv |0\rangle\langle 0| + e^{i\pi/4} |1\rangle\langle 1|$$

$$T|1\rangle = e^{i\pi/4}|1\rangle = \begin{bmatrix} 0 \\ e^{i\pi/4} \end{bmatrix}$$

The T -Gate

- T is unitary

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} \quad \longrightarrow \quad T^\dagger = \left(\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}^* \right)^T = \left(\begin{bmatrix} 1 & 0 \\ 0 & e^{-i\pi/4} \end{bmatrix} \right)^T = \begin{bmatrix} 1 & 0 \\ 0 & e^{-i\pi/4} \end{bmatrix}$$

$$TT^\dagger = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & e^{-i\pi/4} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

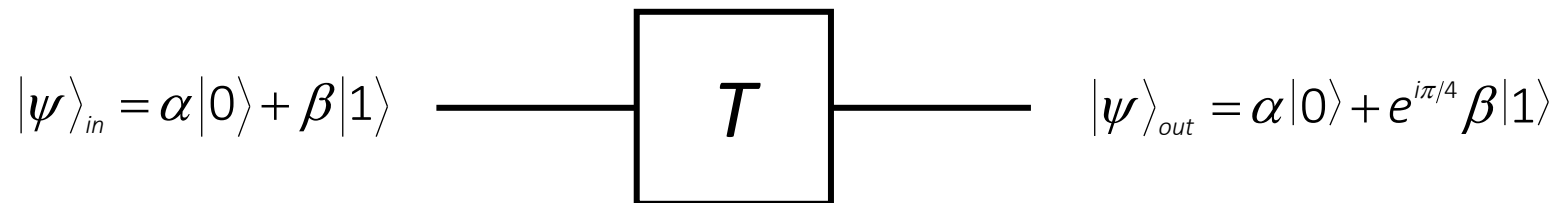
$$T^\dagger T = \begin{bmatrix} 1 & 0 \\ 0 & e^{-i\pi/4} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$\longrightarrow TT^\dagger = T^\dagger T = I$$

The T -Gate

- When acting on $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $|\alpha|^2 + |\beta|^2 = 1$

$$T \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ e^{i\pi/4} \beta \end{bmatrix} = \alpha|0\rangle + e^{i\pi/4} \beta|1\rangle$$



The T -Gate

- You might wonder why the T gate is called the $\pi/8$ gate when it is $\pi/4$ that appears in the definition
- The reason is that the gate has historically often been referred to as the $\pi/8$ gate, simply because up to an unimportant global phase, T is equal to a gate which has $\exp(\pm i\pi/8)$ appearing on its diagonals

$$T = e^{i\pi/8} \begin{bmatrix} e^{-i\pi/8} & 0 \\ 0 & e^{i\pi/8} \end{bmatrix}$$

- Nevertheless, the nomenclature is in some respects rather unfortunate, and we often refer to this gate as the T gate

The Hadamard H -Gate

- This gate, turns $|0\rangle$ into $|+\rangle$, and $|1\rangle$ into $|-\rangle$

$$\left. \begin{aligned} H|0\rangle &= |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ H|1\rangle &= |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \end{aligned} \right\} \longrightarrow H|x\rangle = \frac{|0\rangle + (-1)^x |1\rangle}{\sqrt{2}} \quad \forall x \in \{0,1\}$$

compact form



$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \equiv \frac{1}{\sqrt{2}} (|0\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 0| - |1\rangle\langle 1|)$$

The Hadamard H -Gate

- H is unitary

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \rightarrow H^\dagger = \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^* \right)^T = \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right)^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = H$$

$$HH^\dagger = H^\dagger H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \rightarrow HH^\dagger = H^\dagger H = I$$

The Hadamard H -Gate

- The application of a Hadamard gate to an arbitrary qubit gives the following output

$$H \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \alpha + \beta \\ \alpha - \beta \end{bmatrix} = \left(\frac{\alpha + \beta}{\sqrt{2}} \right) |0\rangle + \left(\frac{\alpha - \beta}{\sqrt{2}} \right) |1\rangle$$

$$= \alpha \frac{|0\rangle + |1\rangle}{\sqrt{2}} + \beta \frac{|0\rangle - |1\rangle}{\sqrt{2}} = \alpha |+\rangle + \beta |-\rangle$$

Quantum Interference

- The application of a Hadamard gate to an arbitrary qubit is an example of *quantum interference*
- In the previous slide we have shown that

$$H \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \frac{\alpha + \beta}{\sqrt{2}} \\ \frac{\alpha - \beta}{\sqrt{2}} \end{bmatrix}$$

Quantum Interference

- Thus

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \alpha|0\rangle + \beta|1\rangle \quad \xrightarrow{\quad H \quad} \quad \begin{bmatrix} \frac{\alpha + \beta}{\sqrt{2}} \\ \frac{\alpha - \beta}{\sqrt{2}} \end{bmatrix} = \frac{\alpha + \beta}{\sqrt{2}}|0\rangle + \frac{\alpha - \beta}{\sqrt{2}}|1\rangle$$

- Notice that the probability to obtain $|0\rangle$ upon measurement has been changed as the probability amplitude

$$\alpha \rightarrow \frac{\alpha + \beta}{\sqrt{2}}$$

while the probability to find $|1\rangle$ has been changed as the probability amplitude

$$\beta \rightarrow \frac{\alpha - \beta}{\sqrt{2}}$$

Quantum Interference

- Now, let's look at the following scenario with $\alpha = \beta = 1/\sqrt{2}$

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \xrightarrow{H} \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

- With the Hadamard transformation of the state $|+\rangle$ we have the following:
 - *Positive interference* with regard to the basis state $|0\rangle$. The two amplitudes add to increase the probability of finding 0 upon measurement. In fact in this case it goes to unity meaning we are certain to find 0.
 - *Negative interference* with regard to the basis state $|1\rangle$. We go from a state where there was a *50% chance of finding 1* upon measurement to one where there is *no chance of finding 1* upon measurement.

Relative Phase

- Obviously, both $|+\rangle$ and $|-\rangle$ will have identical measurement probabilities; if we have 1000 electrons in spin state $|+\rangle$ and 1000 in state $|-\rangle$, a measurement of all of them will throw about $|1/\sqrt{2}|^2 \times 1000 = 500$ into state $|0\rangle$ and $|1/\sqrt{2}|^2 \times 1000 = 500$ into state $|1\rangle$
- *However, two states that have the same measurement probabilities are not necessarily the same state*

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Relative Phase

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \text{---} \quad \boxed{H} \quad \text{---} \quad |0\rangle$$

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad \text{---} \quad \boxed{H} \quad \text{---} \quad |1\rangle$$

- If we give $|+\rangle$ and $|-\rangle$ as input to the H gate we obtain as output $|0\rangle$ and $|1\rangle$ states respectively, i.e. states which give rise to physically observable differences in measurement statistics
- Therefore it is not possible to regard these states ($|+\rangle$ and $|-\rangle$) as physically equivalent, as we do with states differing by a global phase factor

Global Phase

The input states $|+\rangle$ and $e^{i\theta}|+\rangle$ are physically equivalent and so are the output states $|0\rangle$ and $e^{i\theta}|0\rangle$

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \longrightarrow \boxed{H} \longrightarrow |0\rangle \qquad H|+\rangle = |0\rangle$$

$$e^{i\theta}|+\rangle = e^{i\theta} \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \longrightarrow \boxed{H} \longrightarrow e^{i\theta}|0\rangle \qquad H(e^{i\theta}|+\rangle) = e^{i\theta} H \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) = e^{i\theta}|0\rangle$$

It follows from the H linearity

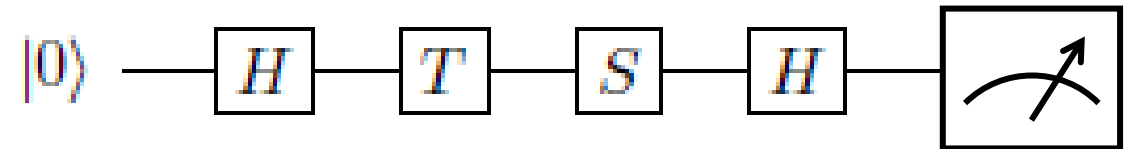
Special 1-Qubit Gates

Gate	Action on Computational Basis	Matrix Representation
Identity	$I 0\rangle = 0\rangle$ $I 1\rangle = 1\rangle$	$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
Pauli X	$X 0\rangle = 1\rangle$ $X 1\rangle = 0\rangle$	$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Pauli Y	$Y 0\rangle = i 1\rangle$ $Y 1\rangle = -i 0\rangle$	$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
Pauli Z	$Z 0\rangle = 0\rangle$ $Z 1\rangle = - 1\rangle$	$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Phase S	$S 0\rangle = 0\rangle$ $S 1\rangle = i 1\rangle$	$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$
T	$T 0\rangle = 0\rangle$ $T 1\rangle = e^{i\pi/4} 1\rangle$	$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$
Hadamard H	$H 0\rangle = \frac{1}{\sqrt{2}}(0\rangle + 1\rangle)$ $H 1\rangle = \frac{1}{\sqrt{2}}(0\rangle - 1\rangle)$	$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Combination of Quantum Gates

- We can combine these quantum gates to create all sorts of states

$$\begin{aligned} HSTH|0\rangle &= HST \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \\ &= HS \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi/4}|1\rangle) \\ &= H \frac{1}{\sqrt{2}} (|0\rangle + e^{i3\pi/4}|1\rangle) \\ &= \frac{1}{\sqrt{2}} \left[\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) + e^{i3\pi/4} \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \right] \\ &= \frac{1}{2} \left[(1 + e^{i3\pi/4}) |0\rangle + (1 - e^{i3\pi/4}) |1\rangle \right], \end{aligned}$$

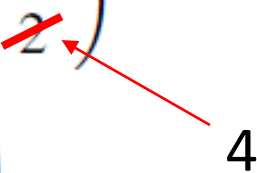


Gates operate from *left-to-right*, but operator algebra moves from *right to-left*. When translating a circuit into a product of matrices, we must reverse the order. So *stay alert!*

where in the third line, we used $ie^{i\pi/4} = e^{i\pi/2}e^{i\pi/4} = e^{i3\pi/4}$.

Combination of Quantum Gates

- Then, if we measure this qubit in the Z-basis $\{|0\rangle, |1\rangle\}$ the probability of getting $|0\rangle$ is

$$\begin{aligned} \left| \frac{1}{2} \left(1 + e^{i3\pi/4} \right) \right|^2 &= \frac{1}{2} \left(1 + e^{i3\pi/4} \right) \frac{1}{2} \left(1 + e^{-i3\pi/4} \right) \\ &= \frac{1}{4} \left(1 + e^{-i3\pi/4} + e^{i3\pi/4} + e^0 \right) \\ &= \frac{1}{4} \left(2 + 2 \cos \frac{3\pi}{2} \right) \\ &= \frac{1}{2} \left(1 - \frac{\sqrt{2}}{2} \right) \\ &\approx 0.146, \end{aligned}$$


where to go from the second to the third line, we used Euler's formula

$$e^{i\vartheta} + e^{-i\vartheta} = 2 \cos \vartheta$$

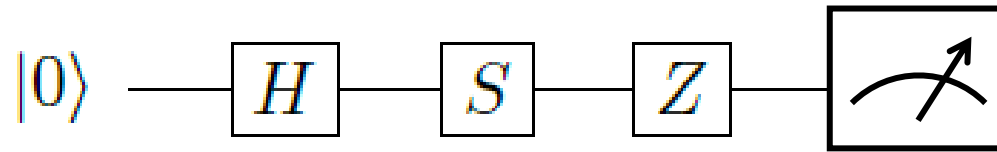
Combination of Quantum Gates

- Similarly, the probability of getting $|1\rangle$ is

$$\left| \frac{1}{2} \left(1 - e^{i3\pi/4} \right) \right|^2 = \frac{1}{4} \left(2 - 2 \cos \frac{3\pi}{2} \right) = \frac{1}{2} \left(1 + \frac{\sqrt{2}}{2} \right) \approx 0.854.$$

Combination of Quantum Gates

- Another combination of quantum gates.



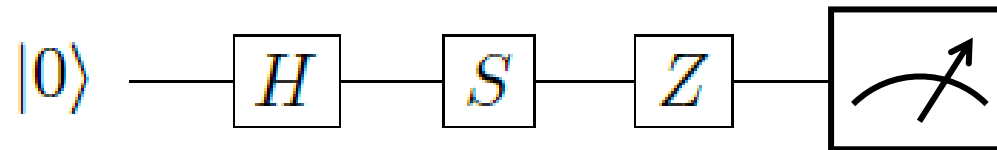
- So we start with a single qubit in the $|0\rangle$ state and apply a Hadamard gate H to it, followed by a phase gate S , and finally a Z gate
- The state at the output of the Z gate, will be $ZSH|0\rangle$

Combination of Quantum Gates

- $ZSH|0\rangle$ results in the state,

$$\begin{aligned} ZSH|0\rangle &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix} = \frac{1}{\sqrt{2}} [|0\rangle - i|1\rangle] \end{aligned}$$

and if we measure the qubit in the Z-basis, we get $|0\rangle$ or $|1\rangle$
with equal probability



Rotation About x-, y-, and z-Axes

- **Question:** what is the most general kind of quantum gate for a single qubit?
- To address this, we must first introduce the family of quantum gates that perform *rotations* about *the three mutually perpendicular axes* of the *Bloch sphere*
- A single qubit state is represented by a point on the surface of the Bloch sphere

Rotation About x-, y-, and z-Axes

- The effect of a single qubit gate that acts in this state is to map it to some other point on the Bloch sphere
- The gates that rotate states around the x-, y-, and z-axes are of special significance since we will be able *to decompose an arbitrary 1-qubit quantum gate into sequences of such rotation gates*
- Any point on the surface of the Bloch sphere can be specified using its (x, y, z) coordinates or, equivalently, its (r, ϑ, ϕ) coordinates (let's ignore the global phase for now)

Rotation About x-, y-, and z-Axes

- These two coordinate systems are related via the equations:

$$x = r \sin(\vartheta) \cos(\phi)$$

$$y = r \sin(\vartheta) \sin(\phi)$$

$$z = r \cos(\vartheta)$$

- So what are the quantum gates that rotate this state about the x-, y-, or z-axes?
- We claim that these gates can be built from the **Pauli X, Y, Z, matrices**, and the **fourth Pauli matrix, I**, can be used to achieve a *global overall phase shift*

Rotation About x-, y-, and z-Axes

Let's define the following unitary matrices

$$R_x(\alpha) = \exp(-i\alpha X/2) = \begin{bmatrix} \cos\left(\frac{\alpha}{2}\right) & -i\sin\left(\frac{\alpha}{2}\right) \\ -i\sin\left(\frac{\alpha}{2}\right) & \cos\left(\frac{\alpha}{2}\right) \end{bmatrix}$$

$$R_z(\alpha) = \exp(-i\alpha Z/2) = \begin{bmatrix} e^{-i\alpha/2} & 0 \\ 0 & e^{i\alpha/2} \end{bmatrix}$$

$$R_y(\alpha) = \exp(-i\alpha Y/2) = \begin{bmatrix} \cos\left(\frac{\alpha}{2}\right) & -\sin\left(\frac{\alpha}{2}\right) \\ \sin\left(\frac{\alpha}{2}\right) & \cos\left(\frac{\alpha}{2}\right) \end{bmatrix}$$

$$Ph(\delta) = e^{i\delta} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Rotation About x-, y-, and z-Axes

To prove (in the above expressions for the rotations) that the exponentials are equivalent to the matrices, you have to prove first that, given a real number x and an A matrix such that $A^2 = -I$ the following relation holds

$$e^{iAx} = \cos(x)I + i\sin(x)A$$

Rotation About x-, y-, and z-Axes

- Consider the gate $R_z(\alpha)$
- Let's see how this gate transforms an arbitrary single qubit

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right)|1\rangle \quad \longrightarrow$$

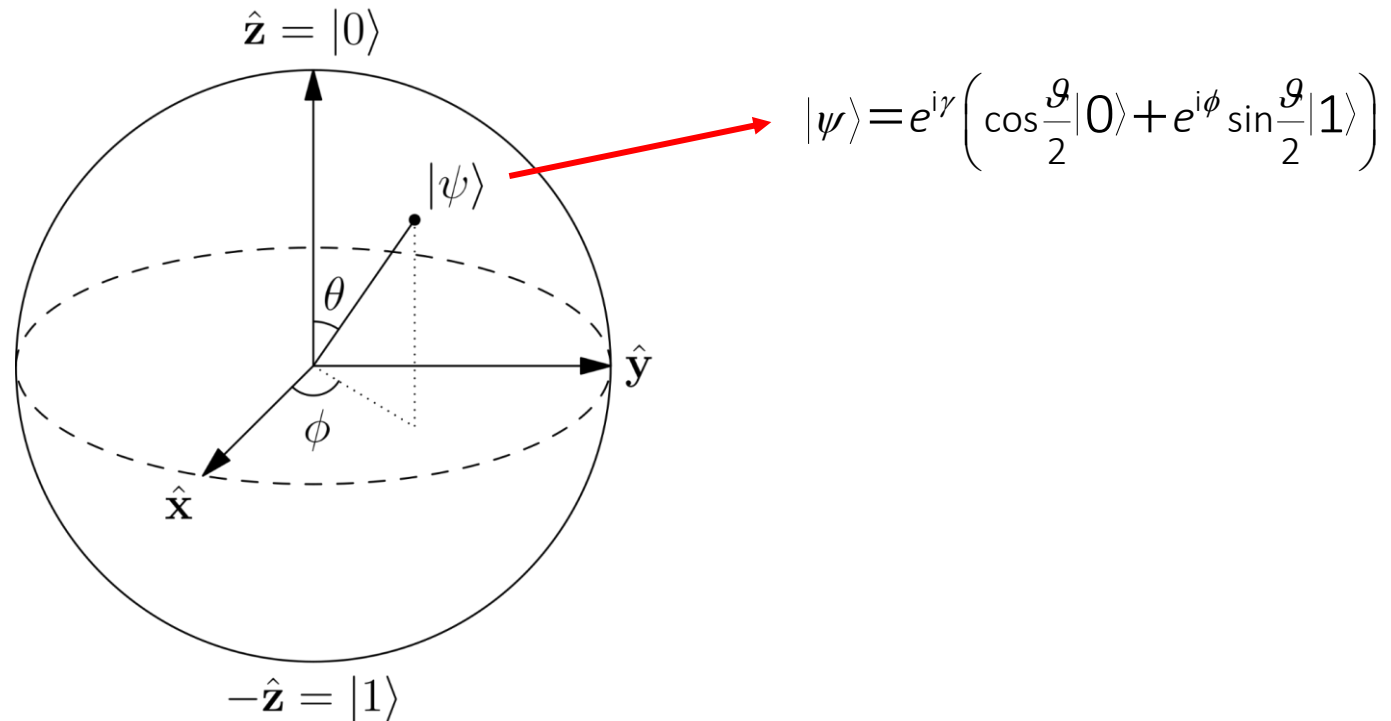
$$R_z(\alpha)|\psi\rangle = \begin{bmatrix} e^{-i\alpha/2} & 0 \\ 0 & e^{i\alpha/2} \end{bmatrix} \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) \\ e^{i\phi} \sin\left(\frac{\theta}{2}\right) \end{bmatrix} = \begin{bmatrix} e^{-i\alpha/2} \cos\left(\frac{\theta}{2}\right) \\ e^{i\alpha/2} e^{i\phi} \sin\left(\frac{\theta}{2}\right) \end{bmatrix} = e^{-i\alpha/2} \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\alpha/2} e^{i\phi} \sin\left(\frac{\theta}{2}\right)|1\rangle$$

Rotation About x-, y-, and z-Axes

From the above slide →

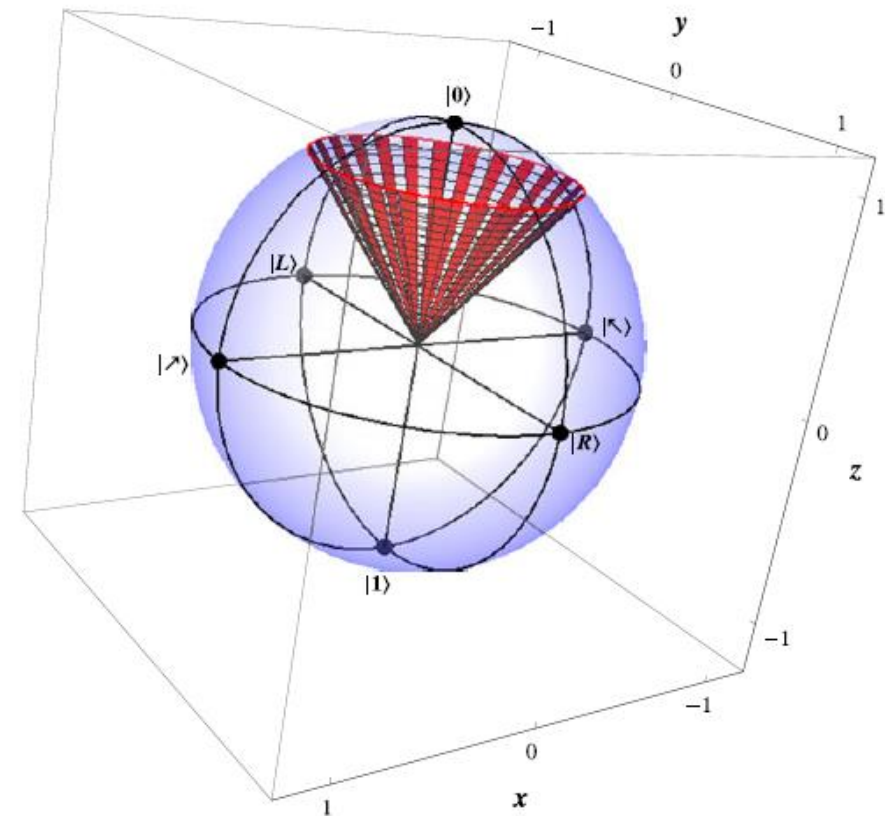
$$R_z(\alpha)|\psi\rangle = e^{-i\alpha/2} \left(\cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i(\phi+\alpha)} \sin\left(\frac{\theta}{2}\right)|1\rangle \right) \equiv \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i(\phi+\alpha)} \sin\left(\frac{\theta}{2}\right)|1\rangle$$

where $\equiv$ is to be read as
“equal up to an unimportant
arbitrary *global phase factor*”



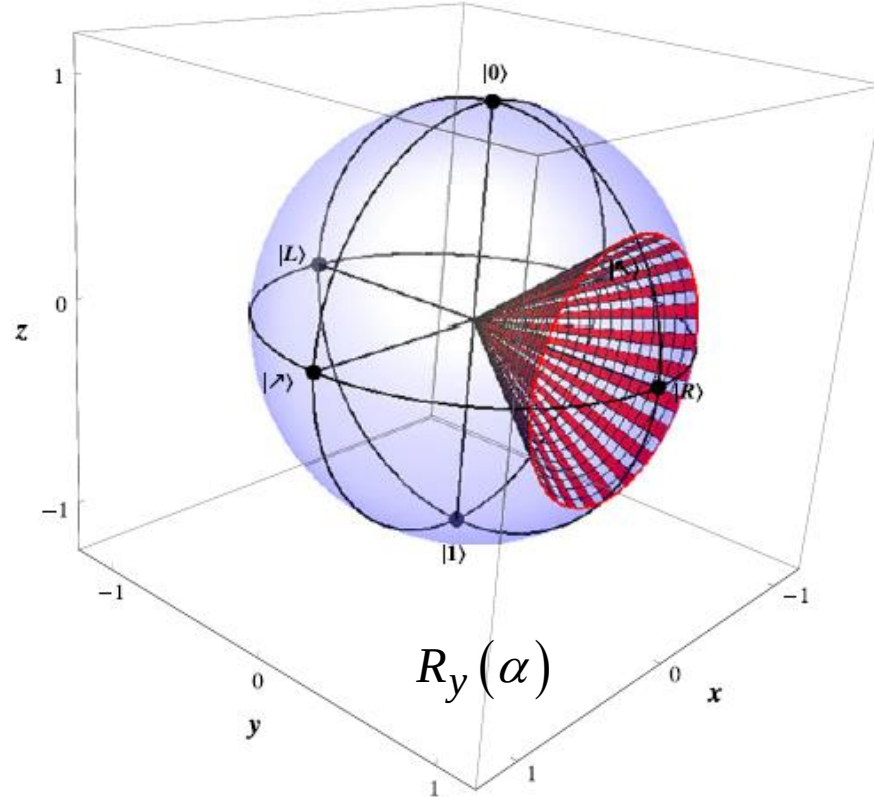
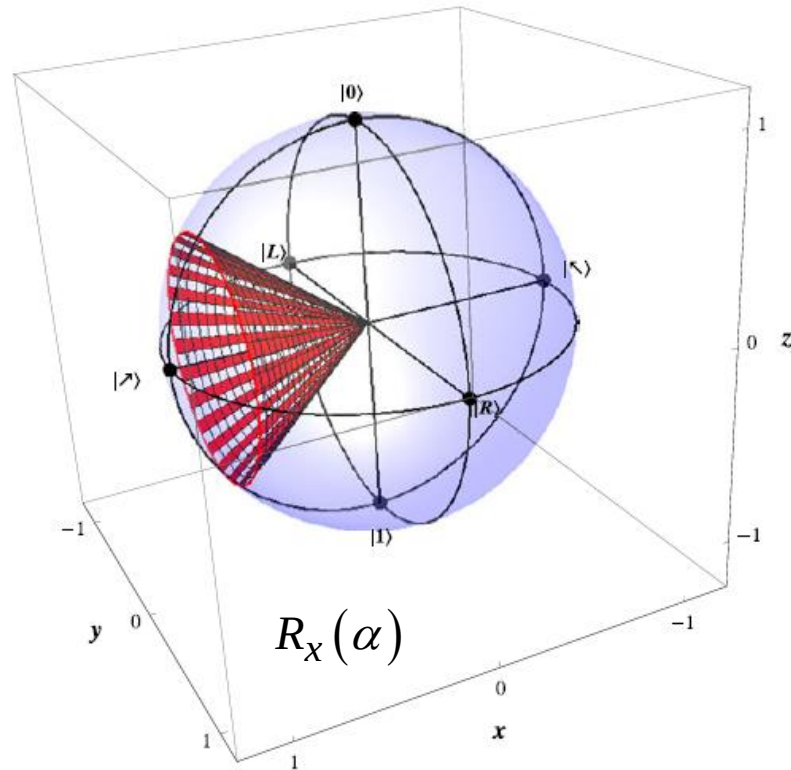
Rotation About x-, y-, and z-Axes

- Hence the action of the $R_z(\alpha)$ gate on $|\psi\rangle$ has been to advance the angle ϕ by α and hence rotate the state about the z-axis through angle α
- This is why we call $R_z(\alpha)$ a z-rotation gate



Rotation About x-, y-, and z-Axes

- We leave it to the exercises for you to prove that $R_x(\alpha)$ and $R_y(\alpha)$ rotate the state about the x- and y-axes respectively



Rotation About x-, y-, and z-Axes

- Rotations on the Bloch sphere do not conform to commonsense intuitions about rotations that we have learned from our experience of the everyday world
- In particular, usually, a rotation of 2π radians (i.e., 360 degrees) of a solid object about any axis, restores that object to its initial orientation

Rotation About x-, y-, and z-Axes

- However, this is not true of rotations on the Bloch sphere! When we rotate a quantum state through 2π on the Bloch sphere we don't return it to its initial state
- Instead we pick up a phase factor

Rotation About x-, y-, and z-Axes

- To see this, let's compute the effect of rotating our arbitrary single qubit state

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right)|1\rangle$$

about the z-axis through 2π radians

$$R_z(2\pi)|\psi\rangle = \begin{bmatrix} e^{-i\pi} & 0 \\ 0 & e^{i\pi} \end{bmatrix} \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) \\ e^{i\phi} \sin\left(\frac{\theta}{2}\right) \end{bmatrix} = \begin{bmatrix} -\cos\left(\frac{\theta}{2}\right) \\ -e^{i\phi} \sin\left(\frac{\theta}{2}\right) \end{bmatrix} = -|\psi\rangle$$

which has an extra overall phase of -1

Rotation About x-, y-, and z-Axes

- To restore a state back to its original form we need to rotate it through 4π on the Bloch sphere
- Have you ever encountered anything like this in your everyday world? You probably think not, but you'd be wrong!
- See the “Dirac’s Belt” or the “Belt Trick” video <https://www.youtube.com/watch?v=Vfh21o-JW9Q>

X From a Rotation Gate

- On the Bloch sphere, it can be shown that X is a rotation of 180° about the x-axis together with *global phase shifts* of 90°

$$R_x(\pi) \cdot Ph\left(\frac{\pi}{2}\right) = \begin{bmatrix} \cos\frac{\pi}{2} & -i\sin\frac{\pi}{2} \\ -i\sin\frac{\pi}{2} & \cos\frac{\pi}{2} \end{bmatrix} e^{i\pi/2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2} \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

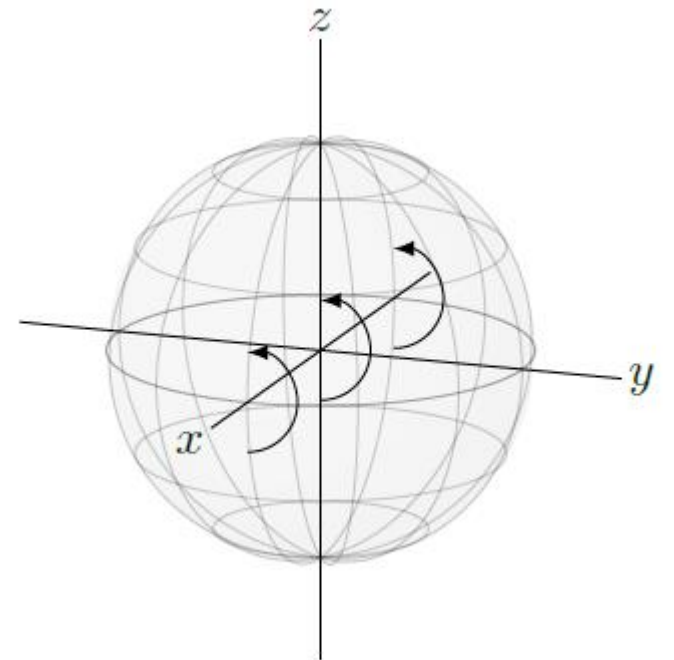
$$= \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} (i) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = X$$

X From a Rotation Gate

- With this rotation in mind, we geometrically see that X causes $|0\rangle$ (the north pole) to rotate to $|1\rangle$ (the south pole), and vice versa
- We also see that $|i\rangle$ and $|-i\rangle$ rotate to each other, whereas $|+\rangle$ and $|-\rangle$ are unchanged
- Note, however, that mathematically

$$X|-\rangle = -|-\rangle \equiv |-\rangle$$

since the global phase does not matter

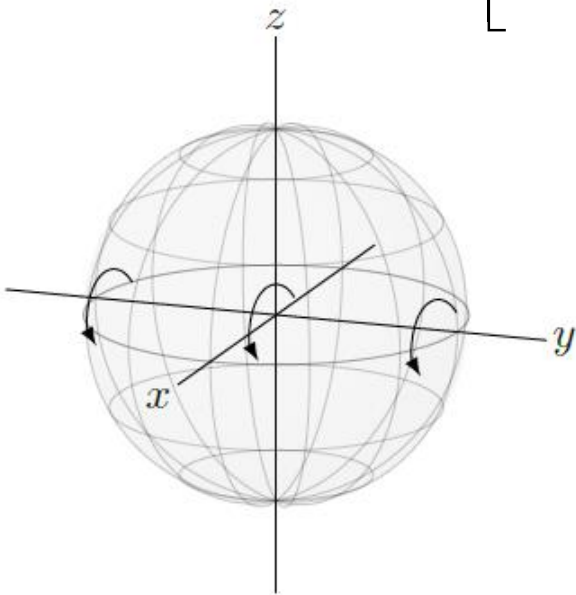


Y From a Rotation Gate

- On the Bloch sphere, it can be shown that Y is a rotation of 180° about the y -axis together with *global phase shifts* of 90°

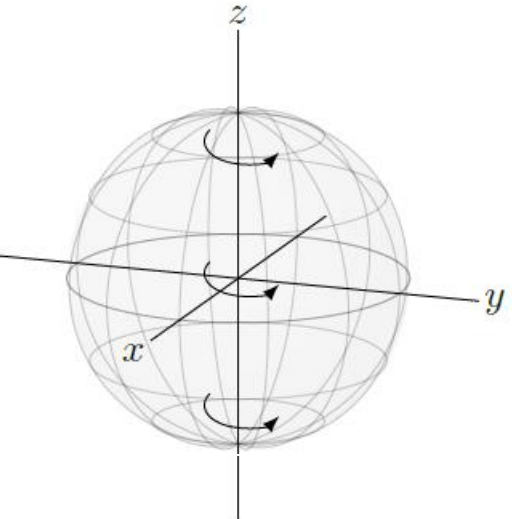
$$R_y(\pi) \cdot Ph\left(\frac{\pi}{2}\right) = \begin{bmatrix} \cos \frac{\pi}{2} & -\sin \frac{\pi}{2} \\ \sin \frac{\pi}{2} & \cos \frac{\pi}{2} \end{bmatrix} e^{i\pi/2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} (i) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = Y$$



Z From a Rotation Gate

- On the Bloch sphere, it can be shown that Z is a rotation of 180° about the z-axis together with *global phase shifts* of 90°

$$\begin{aligned}
 R_Z(\pi) \cdot Ph\left(\frac{\pi}{2}\right) &= \begin{bmatrix} e^{-i\pi/2} & 0 \\ 0 & e^{i\pi/2} \end{bmatrix} e^{i\pi/2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\frac{\pi}{2} - i\sin\frac{\pi}{2} & 0 \\ 0 & \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} \end{bmatrix} \left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2} \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} \left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2} \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} (i) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = Z
 \end{aligned}$$


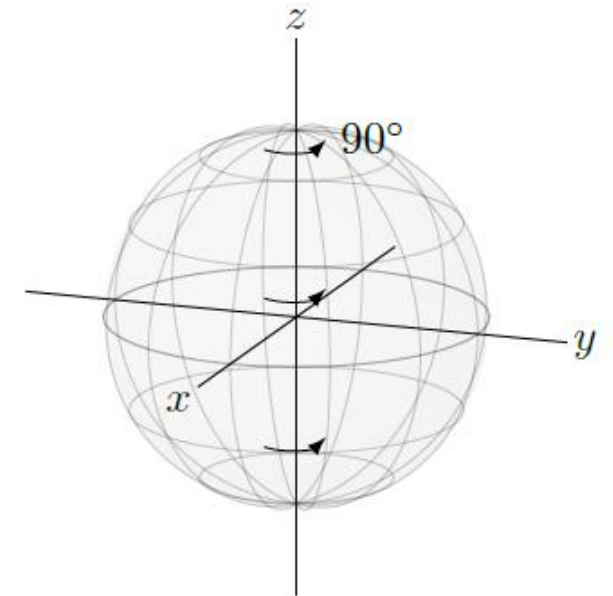
S From a Rotation Gate

- On the Bloch sphere, it can be shown that S is a rotation of 90° about the z -axis together with *global phase shifts* of 45°

$$R_z\left(\frac{\pi}{2}\right) \cdot Ph\left(\frac{\pi}{4}\right) = \begin{bmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} e^{i\pi/8} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\frac{\pi}{4} - i\sin\frac{\pi}{4} & 0 \\ 0 & \cos\frac{\pi}{4} + i\sin\frac{\pi}{4} \end{bmatrix} \left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4} \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1-i & 0 \\ 0 & 1+i \end{bmatrix} \left(\frac{1}{\sqrt{2}} (1+i) \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

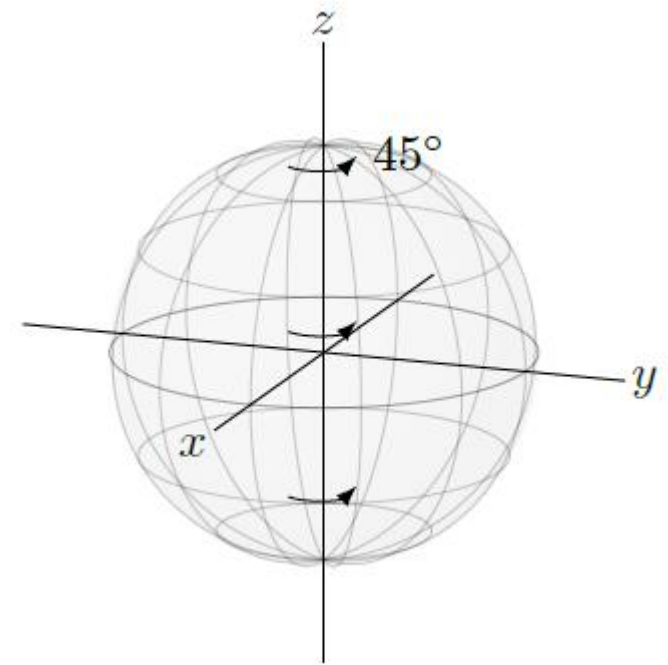
$$= \frac{1}{2} \begin{bmatrix} 1-i & 0 \\ 0 & 1+i \end{bmatrix} \begin{bmatrix} 1+i & 0 \\ 0 & 1+i \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1-i^2 & 0 \\ 0 & (1+i)^2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2i \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} = S$$



T From a Rotation Gate

- On the Bloch sphere, it can be shown that T (also called $\pi/8$ gate) is a rotation of 45° about the z -axis together with *global phase shifts* of $\pi/8$ radians

$$\begin{aligned} R_z\left(\frac{\pi}{4}\right) \cdot Ph\left(\frac{\pi}{8}\right) &= \begin{bmatrix} e^{-i\pi/8} & 0 \\ 0 & e^{i\pi/8} \end{bmatrix} e^{i\pi/8} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} = T \end{aligned}$$



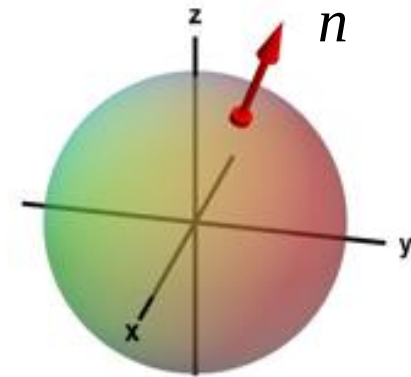
H From Rotation Gates

$$\begin{aligned} R_x(\pi) \cdot R_y\left(\frac{\pi}{2}\right) \cdot Ph\left(\frac{\pi}{2}\right) &= \begin{bmatrix} \cos \frac{\pi}{2} & -i \sin \frac{\pi}{2} \\ -i \sin \frac{\pi}{2} & \cos \frac{\pi}{2} \end{bmatrix} \begin{bmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{bmatrix} e^{i\pi/2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{i}{\sqrt{2}} \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = H \end{aligned}$$

Rotation About an Arbitrary Axis

- If $n = [n_x, n_y, n_z]$ is a real unit vector in three dimensions, then it can be shown that the operator $R_n(\theta)$ rotates the Bloch vector by an angle θ about the n axis, where

$$R_n(\theta) = \exp\left(-i\theta\left(n \cdot \frac{\vec{\sigma}}{2}\right)\right)$$



and $\vec{\sigma}$ denotes the three component vector (X,Y,Z) of Pauli matrices

Rotation About an Arbitrary Axis

- Furthermore, it is not hard to show that $(\vec{n} \cdot \vec{\sigma})^2 = I$, and therefore we can use the special case operator exponential and write

$$\begin{aligned} R_n(\theta) &= \cos\left(\frac{\theta}{2}\right)I - i\sin\left(\frac{\theta}{2}\right)\vec{n} \cdot \vec{\sigma} \\ &= \cos\left(\frac{\theta}{2}\right)I - i\sin\left(\frac{\theta}{2}\right)(n_x X + n_y Y + n_z Z) \end{aligned}$$

Arbitrary Unitary Operator

- It can be shown that *an arbitrary single qubit unitary operator* can be written in the form

$$U = \exp(i\alpha) R_n(\theta)$$

for some real number α and θ , and a real three-dimensional unit vector n



$$U = \exp(i\alpha) R_n(\theta) = \exp(i\alpha) \left[\cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) (n_x X + n_y Y + n_z Z) \right]$$

Arbitrary Unitary Operator

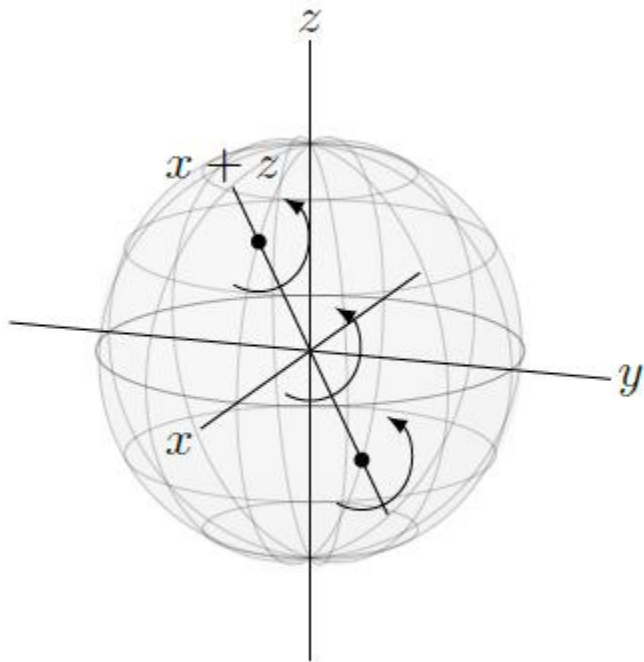
- For example, consider

$$\alpha = \frac{\pi}{2}, \quad \theta = \pi, \quad \text{and} \quad n = \left[\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right] \quad \rightarrow$$

$$\begin{aligned} U &= \exp(i\pi/2) R_n(\pi) = \exp(i\pi/2) \left[\cos\left(\frac{\pi}{2}\right) I - i \sin\left(\frac{\pi}{2}\right) \frac{1}{\sqrt{2}} (X + Z) \right] \\ &= \exp(i\pi/2) \left[\cos\left(\frac{\pi}{2}\right) I - i \sin\left(\frac{\pi}{2}\right) \frac{1}{\sqrt{2}} (X + Z) \right] = \left[\cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) \right] \left[-i \frac{1}{\sqrt{2}} (X + Z) \right] \\ &= \frac{1}{\sqrt{2}} (X + Z) = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = H \end{aligned}$$

H From a Rotation Gate

Thus, on the Bloch sphere, H is a rotation of 180° about the $x+z$ -axis together with *global phase shifts* of 90°



Arbitrary Unitary Operator

- An arbitrary unitary operator on a single qubit can be written in many ways as a combination of rotations, together with *global phase shifts* on the qubit
- For example, we have already seen that

$$X = R_x(\pi) \cdot Ph\left(\frac{\pi}{2}\right) \qquad H = R_x(\pi) \cdot R_y\left(\frac{\pi}{2}\right) \cdot Ph\left(\frac{\pi}{2}\right)$$

- However, we can easily show that

$$X = R_y(\pi) \cdot R_z(\pi) \cdot Ph\left(\frac{\pi}{2}\right) \qquad H = R_y\left(\frac{\pi}{2}\right) \cdot R_z(\pi) \cdot Ph\left(\frac{\pi}{2}\right)$$

Arbitrary Unitary Operator

- The following theorem will be particularly useful in later applications to controlled operations.
- *Theorem: (Z-Y **decomposition for a single qubit**)* Suppose U is a unitary operation on a single qubit. Then there exist real numbers α , β , γ and δ such that

$$U = \exp(i\alpha) R_z(\beta) \cdot R_y(\gamma) \cdot R_z(\delta)$$